

DASE7506 MP1: Architectural Improvements and Experimental Analysis of a Small GPT

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Final test BPB: 1.599273

Abstract

This project improves the course baseline GPT under the fixed WikiText-2 evaluation protocol. Starting from a full-test BPB of **2.1013**, the development process compared feed-forward gating, positional encoding, attention normalization, model capacity, and dropout. The selected model combines SwiGLU, RoPE, head-wise Q/K LayerNorm, width 192, depth 6, and residual dropout 0.10. It achieves **1.5993 test BPB**, a **23.89% reduction** from the baseline. CPU scoring time is 1.65 times the baseline, peak evaluation memory is 2.112 GiB, and uncompressed inference assets occupy approximately 11.82 MiB. The final model satisfies all course resource limits.

1. Experimental Setup and Model Comparison

1.1 Evaluation Protocol

The supplied WikiText-2 splits, train-fitted BPE-2048 tokenizer, evaluator, and independent 256-token causal-window protocol were unchanged. Models were selected on validation and evaluated on the full test split only after freezing; scoring used CPU FP32. Limits are 5× baseline CPU time, 4 GiB peak RAM, and 64 MiB uncompressed inference assets.

1.2 Baseline and Final Model

The baseline and final model are decoder-only Transformers with causal attention, residual connections, LayerNorm, and tied token embeddings. Their main differences are:

Component	Baseline	Final model
Width / depth	128 / 4	192 / 6
Heads / head dimension	4 / 32	6 / 32
Feed-forward network	4×width GELU MLP	SwiGLU, hidden size 512
Position encoding	Learned absolute	RoPE, $\theta=10000$
Q/K normalization	None	Head-wise LayerNorm
Residual dropout	0	0.10
Parameters	1.088M	3.058M
Training budget	9.83M targets	39.32M targets
Full-test BPB	2.1013	1.5993

2. Model Development and Comparative Experiments

This chapter describes each method used, the corresponding model or training change, and the evidence for retaining it. The cumulative sequence uses one training program, seed, batch size, optimizer, and learning-rate schedule. Parameter screens answer separate questions and are compared only within matched groups.

2.1 Cumulative Development Path

Each stage starts from scratch, adds one change to the preceding architecture, and uses 1,200 steps through E; F extends training to 4,800 steps and G adds dropout 0.10. F and G reuse the matched long-run dropout runs.

Stage	Cumulative change	Validation BPB	ΔBPB from prior stage
0	Baseline, width 128, depth 4	2.0842	-
A	Add SwiGLU	2.0238	-0.0604
B	Add RoPE, $\theta=10000$	1.8434	-0.1804
C	Add head-wise Q/K LayerNorm	1.8322	-0.0111
D	Expand width to 192, heads to 6, FFN hidden size to 512	1.7510	-0.0813
E	Increase depth from 4 to 6	1.7272	-0.0238
F	Extend training from 1,200 to 4,800 steps	1.6570	-0.0702
G	Add residual dropout 0.10	1.5752	-0.0818

Width expansion also changes head count and FFN size to keep head dimension 32, so it measures joint capacity scaling, not width alone. Each ΔBPB is conditional on the preceding stage and one seed. Stage 0 is a fresh baseline rerun (2.0842 validation BPB); the separately frozen baseline scored 2.0711 validation and 2.1013 test BPB.

2.2 SwiGLU Feed-Forward Network

Method and change. SwiGLU replaces the baseline GELU MLP with two input projections. In `code/student.py`, the SiLU-activated gate is multiplied element-wise by the value path, then projected back to the model width:

```
out(SiLU(gate(x)) × value(x)).
```

Evidence. With seed 17, width 128, depth 4, learned positions, no dropout, batch size 16, 2,400 updates, and 9.83M targets, GELU scores 1.9918 BPB with 1.088M parameters, whereas SwiGLU scores **1.9288** with 1.090M. The 0.0630 reduction costs only about 1,536 parameters, so SwiGLU is retained.

2.3 Rotary Position Embedding

Method and change. RoPE replaces learned absolute position embeddings. RotaryEmbedding precomputes sine/cosine values and rotates Q/K in every attention head before their dot product, making attention sensitive to relative displacement. At width 128 and context 256, removing the learned position table also removes 32,768 parameters.

Evidence. In the paired seed-17 SwiGLU experiment with width 128, depth 4, batch size 16, and 2,400 updates, learned positions score 2.0222 BPB and RoPE scores **1.7837**, a 0.2384 reduction and the largest observed component gain. This earlier implementation revision is interpreted separately from the GELU/SwiGLU confirmation.

2.4 Head-Wise Q/K LayerNorm

Method and change. Separate `nn.LayerNorm(head_dim)` modules normalize Q and K within each attention head before RoPE. This controls their scale and reduces excessively sharp attention logits.

Evidence. In the $\theta=1000$ paired experiment, disabling Q/K LayerNorm scores 1.7605 BPB and enabling it scores **1.7510**, a 0.0095 gain. The method is retained, although this single-seed benefit is smaller and less certain than the SwiGLU and RoPE gains.

2.5 Architecture Parameter Search

After selecting the main components, head dimension, width, and depth were screened to balance validation BPB against parameter count and CPU training time. The selected settings are head dimension 32, width 192, and depth 6.

All runs in the table use SwiGLU, RoPE, Q/K LayerNorm, LayerNorm, zero dropout, vocabulary size 2,048, context length 256, seed 17, batch size 16, and 2,400 updates. Each candidate is listed separately so that the architectural settings and resource trade-offs are explicit.

Search	Width	Depth	Heads	Head dim.	FFN hidden	RoPE θ	Parameters	CPU train time (s)	Validation BPB	Selected
Head dimension	192	6	12	16	512	1,000	3,057,408	798.5	1.6968	
Head dimension	192	6	6	32	512	1,000	3,057,792	721.5	1.6888	Yes
Head dimension	192	6	4	48	512	1,000	3,058,176	958.6	1.6899	
Head dimension	192	6	3	64	512	1,000	3,058,560	653.0	1.6886	
Width	160	6	5	32	432	1,000	2,195,008	613.0	1.7122	
Width	192	6	6	32	512	1,000	3,057,792	721.5	1.6888	Yes
Width	224	6	7	32	600	1,000	4,094,144	897.2	1.6780	
Depth	192	4	6	32	512	10,000	2,169,728	405.7	1.7138	
Depth	192	6	6	32	512	10,000	3,057,792	603.1	1.6912	Yes
Depth	192	8	6	32	512	10,000	3,945,856	823.4	1.6861	

Within the head-dimension sweep, 64 improves on 32 by only 0.0002 BPB, so 32 is retained for the simpler six-head layout. Width 224 adds 1,036,352 parameters and 175.7 seconds relative to width 192 for a 0.0108 reduction. Depth 8 adds 888,064 parameters and 220.3 seconds relative to depth 6 for a 0.0051 reduction. The head-dimension and width screens use $\theta=1000$, whereas the depth screen and final model use $\theta=10000$; absolute BPB is therefore compared only within each search group.

2.6 Dropout and Long-Run Training

Without dropout, BPB worsens from 1.6504 at step 3,600 to 1.6570 at step 4,800. Residual dropout is therefore applied after attention and SwiGLU output projections, before residual additions; it is training-only and adds no inference parameters.

Dropout	Validation BPB at step 4,800
0	1.6570
0.05	1.6095
0.10	1.5752

Validation BPB over 4,800 steps

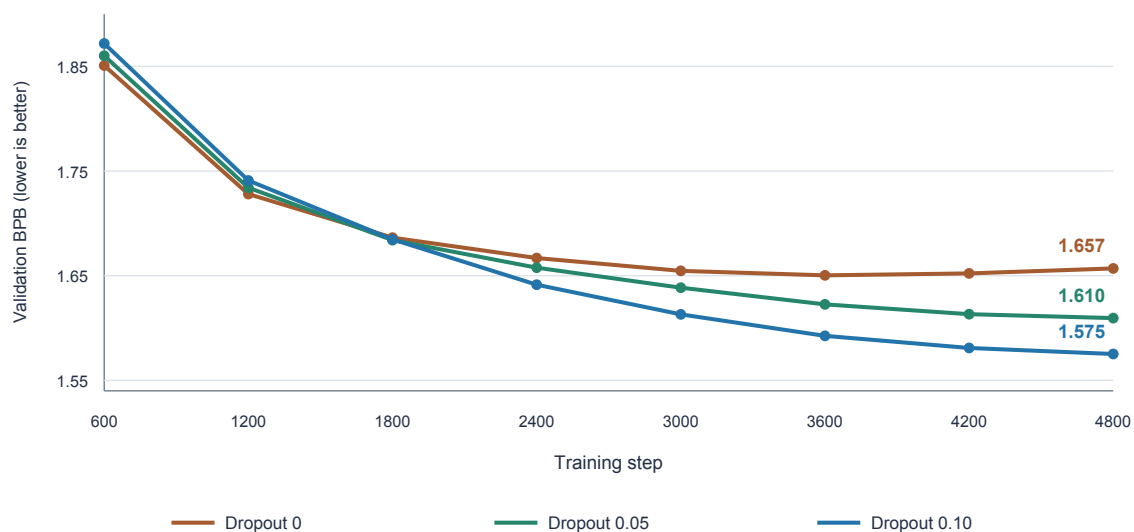


Figure 1. Validation BPB in matched long-run dropout experiments.

Runs share seed 17, batch size 32, final architecture, 4,800 updates, 200-step warmup, and cosine decay. Dropout 0.10 is best among tested values. The recipe uses AdamW (beta1=0.9, beta2=0.999; peak LR 0.001); BPB falls from 1.8719 at step 600 to 1.5752 at 4,800, so this is a budget-selected point, not demonstrated convergence.

3. Ablation Study

A strict ablation retrains each one-component variant from scratch using `student_ablation.py`, seed 17, batch size 32, 4,800 steps, the same AdamW/warmup/cosine recipe, and validation split. The full control reproduces **1.5752 BPB**; 1.5993 is the frozen test score and is not used for selection. Here ΔBPB = ablated - full; positive values indicate degradation.

Final-configuration variant	Parameters	Validation BPB	ΔBPB
Full model	3,057,792	1.5752	-
Replace SwiGLU with GELU MLP	3,063,552	1.6262	+0.0510
Replace RoPE with learned absolute positions	3,106,944	1.6312	+0.0560
Disable head-wise Q/K LayerNorm	3,057,024	1.6115	+0.0363
Set dropout from 0.10 to 0	3,057,792	1.6570	+0.0818

Removing or replacing each of the four components increases validation BPB relative to the full model. Replacing SwiGLU, replacing RoPE, disabling Q/K LayerNorm, and removing dropout increase BPB by 0.0510, 0.0560, 0.0363, and 0.0818, respectively. These controlled ablations demonstrate that all four optimization measures contribute to the final model's performance under the tested configuration, with dropout producing the largest observed contribution.

GELU and learned positions differ from the full model by +5,760 and +49,152 parameters, respectively, so those comparisons are approximate rather than exact parameter matches. The experiments use one seed and therefore cannot quantify variability across seeds or establish that the same gains hold for other configurations.

4. Final Results and Critical Analysis

4.1 Test Result and Resource Compliance

After validation selection, the model and checkpoint were frozen before full-test evaluation.

Metric	Baseline	Frozen model	Requirement or interpretation
Full-test BPB	2.1013	1.5993	23.89% reduction
CPU FP32 scoring time	5.844 s	9.624 s	≤ 29.220 s
Peak evaluation RSS	-	2.112 GiB	≤ 4 GiB
Uncompressed inference assets	-	11.82 MiB	≤ 64 MiB

The model meets all limits. A same-hash replay scored in 9.115 s. Peak RAM is process RSS; the evaluator JSON's `peak_alllocated_gb=0` means no GPU allocation, not zero CPU RAM. The 11.82 MiB inference bundle includes checkpoint, implementation, config, and tokenizer, excluding logs. Final training/replay took about 2,636/2,573 s; 45 runs are logged (41 timed runs total about 13.18 CPU hours).

4.2 Critical Analysis

Architecture-Training Interaction

The preferred dropout depends on training horizon: 0.05 at 2,400 steps, 0.10 at 4,800. Component gains may interact and should not be added as independent effects.

Marginal Gains and Resource Cost

Scaling has diminishing returns: width 224 and depth 8 improve BPB only slightly for many more parameters and CPU seconds. Width 192/depth 6 therefore balances quality and cost; the final model uses 2.81× baseline parameters and 1.65× scoring time.

Training Budget and Schedule Attribution

BPB still improves at step 4,800, so convergence is unproven. Warmup, cosine decay, dropout, and longer training were combined; without schedule ablations their individual effects cannot be separated.

4.3 Reproducibility

The frozen model uses seed 17 and `student_swiglu_rope_width192_head32_theta10000_qknorm_dropout010.json`. A 4,800-step replay reproduced the checkpoint hash, and run metrics are recorded in `code/run_log.csv`.

From `code/`, reproduce final training with:

```
python train.py --implementation student \
  --config configs/student_swiglu_rope_width192_head32_theta10000_qknorm_dropout010.json \
  --device cpu --threads 4 --seed 17 --steps 4800 \
  --eval-every 600 --save-every 600 \
  --run-dir runs/reproduce-final-dropout010-s4800-s17
```

Evaluate the frozen checkpoint without retraining:

```
python evaluate.py \
  --checkpoint runs/lr-verify-warmup200-cosine-dropout010-s4800-b32-s17/checkpoint.pt \
  --device cpu --threads 4 --precision fp32 --split test
```

The frozen checkpoint is stored at `code/runs/lr-verify-warmup200-cosine-dropout010-s4800-b32-s17/checkpoint.pt`.

Artifact	SHA-256
Frozen checkpoint	04f8d2c00a70d528b2a3b31de3ad3717 cca93c18df616550f69cb59645bacbfa
Matching student.py	47477124058e73a234d60d75af31af1d 80e12d3a89d4cb1c1f831e7ae0ee27e5

5. AI Assistance

AI tools helped propose improvements, plan controlled and ablation experiments, implement and review code/configurations, organize metrics and resource measurements, and draft this report. All reported values come from saved runs and were checked against project logs and artifacts.